

RESULTS: KL REGULARIZATION AGAINST AN EMA REFERENCE

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1. SETUP

We train Qwen3-4B-Instruct-2507 with TBA on Countdown using asynchronous rollouts ($\Delta = 10$), $\alpha = 0.9$ for the EMA reference, $\beta = 0.005$, $\text{lr} = 10^{-6}$, $K = 16$ samples per problem, sequence length 4096, 720 training steps, evaluation every 20 steps.

Per-token KL.. Two choices for the per-token KL contribution to the advantage:

- *Exact KL*: $\log T_n(y) - \log E_n(y)$, where E_n is the EMA reference policy.
- *Approx KL*: $\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$, the first-order surrogate derived in `main.tex` (eq. 2.10). This avoids a forward pass through E_n at training time.

Centering. For each rollout, the per-token KL is summed over masked tokens to give a per-rollout KL term $K^{(i)}$. The advantage is then adjusted by

$$(1) \quad \text{advantage}^{(i)} \leftarrow \text{advantage}^{(i)} - \beta \cdot (K^{(i)} - \bar{K}),$$

where $\bar{K}$ is a centering term. Two choices:

- *K-centered*: $\bar{K}$ is the mean of $\{K^{(j)}\}_{j=1}^K$ over the K rollouts that share a problem.
- *No centering*: $\bar{K} \equiv 0$, so $\text{advantage}^{(i)} \leftarrow \text{advantage}^{(i)} - \beta K^{(i)}$ directly.

2. 2×2 MATRIX: EMA REFERENCE

Crossing the two per-token KL choices with the two centering choices yields four cells:

Cell	Per-sample KL formula	Centering
Exact, K-centered	$\log T_n(y) - \log E_n(y)$	$\bar{K}$ from same formula
Exact, no centering	$\log T_n(y) - \log E_n(y)$	$\bar{K} = 0$
Approx, K-centered	$\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$	$\bar{K}$ from same formula
Approx, no centering	$\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$	$\bar{K} = 0$

Cell	Peak eval (step)	Final eval
Exact, K-centered	0.789 (240)	0.656
Exact, no centering	0.827 (620)	0.821
Approx, K-centered	0.794 (380)	0.548
Approx, no centering	0.831 (520)	0.819

The headline observations:

- (1) Within either KL choice, removing the centering term strictly helps.
- (2) The approx-KL row tracks the exact-KL row closely: peaks differ by 0.005 and finals by 0.002 in the no-centering pair. This is the empirical justification for substituting the first-order surrogate (no extra forward) for the exact EMA KL.
- (3) The K-centered cells exhibit the same kind of late-run instability under both per-token KL choices, suggesting the failure mode is centering-related rather than KL-formula-related.

3. RESET REFERENCE (BASELINE)

Before the EMA work, the same kinds of variants were tested with a reset reference policy (hard copy from the training policy every 50 rollout steps). Reset-ref runs use the *exact* KL only (the approximation requires an EMA), and explore three choices for where to obtain $\log T_n$

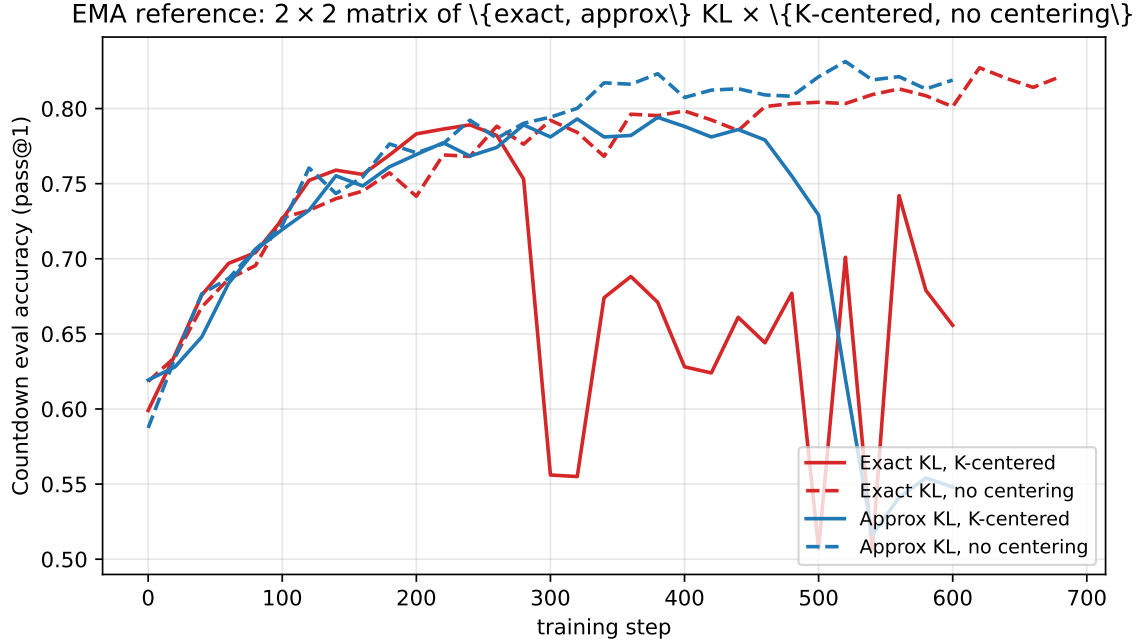


FIGURE 1. Eval accuracy on Countdown for the four cells of the EMA-reference matrix. Red: exact KL. Blue: approx KL. Solid: K-centered. Dashed: no centering. The two no-centering cells (dashed) plateau above 0.80 and peak above 0.82; the two K-centered cells (solid) peak earlier and degrade after step ≈ 300 .

in the per-sample term and in the K-group mean. Naming below: “inf-mean” / “train-mean” indicates which source is used for $\log T_n$ in the K-group mean.

Run	Per-sample KL ($\log T_n$ from)	K-group mean ($\log T_n$ from)
Exact KL, K-centered (inf-mean)	live train forward	inference snapshot
Exact KL, K-centered (train-mean)	live train forward	live train forward
Inf-snapshot KL, K-centered (inf-mean)	inference snapshot	inference snapshot

Each of the three is run with and without importance sampling, for six total curves in Figure 2.

Run	Peak eval (step)	Final eval
Exact KL, K-centered (inf-mean)	0.836 (700)	0.836
Exact KL, K-centered (train-mean)	0.798 (480)	0.122
Inf-snapshot KL, K-centered (inf-mean)	0.752 (220)	0.070
Exact KL, K-centered (inf-mean) no IS	0.744 (680)	0.664
Exact KL, K-centered (train-mean) no IS	0.708 (40)	0.557
Inf-snapshot KL, K-centered (inf-mean) no IS	0.613 (20)	0.000

The train-mean centering instability under reset reference mirrors the K-centered instability in the EMA matrix.

4. APPROXIMATION ACCURACY

We track the per-token error of the first-order surrogate against the exact EMA KL inside `tba_loss`: **MAE** (mean absolute error over masked tokens) and **bias** (signed mean). Figure 3 compares the two no-centering cells (where the loss path differs but both have the same telemetry).

MAE stays in the low 10^{-2} band throughout training, and the signed bias is at most a few parts in 10^3 . The approximation is empirically faithful, regardless of whether it is used in the loss.

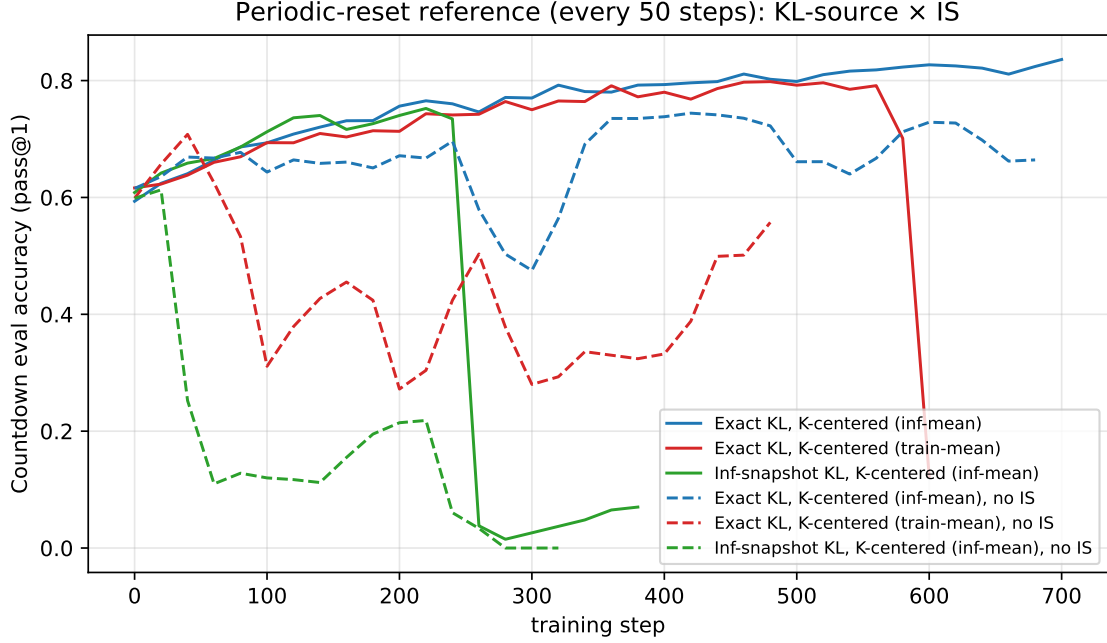


FIGURE 2. Eval accuracy on Countdown with periodic-reset reference. Both *Exact KL, K-centered (train-mean)* variants (red) collapse late in training; the *Inf-snapshot* variant (green) without IS gets stuck at zero; only the *Exact KL, K-centered (inf-mean)* variants (blue) trained stably.

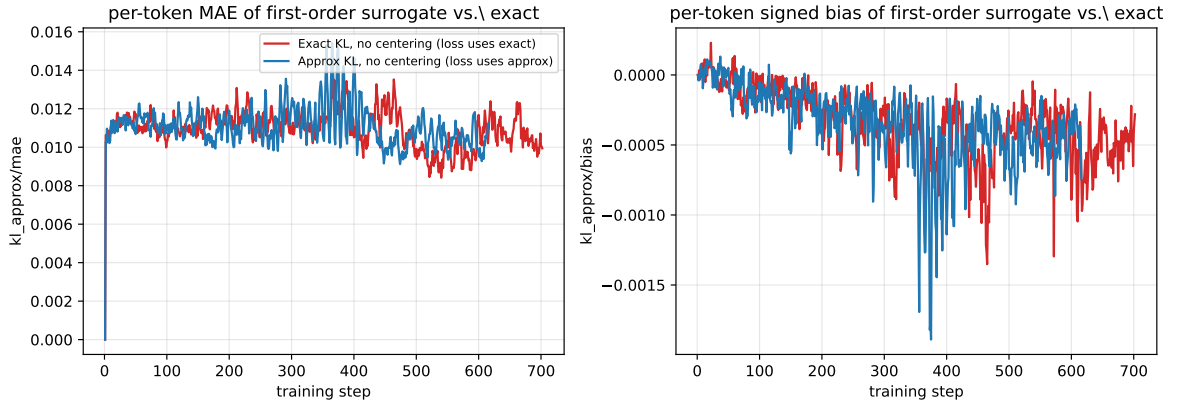


FIGURE 3. Per-token MAE (left) and signed bias (right) of the first-order surrogate $\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$ against the exact $\log T_n(y) - \log E_n(y)$, computed over masked tokens within each rollout. Both no-centering cells: red uses the exact KL in the loss; blue uses the approx KL in the loss.

5. CONCLUSION

Removing the K-group centering is the dominant lever: both EMA exact and EMA approx benefit. With centering off, the cheap first-order surrogate matches the exact EMA KL within 0.005 on peak eval and 0.002 on final eval, so it is a viable drop-in replacement that saves a forward pass through the reference model.